

Introduction to Non-Parametric Methods

Presentation Outline

1 Example 1

2 Example 2

Data Description and Variables

Educational researchers randomly assigned 28 9th-grade students in Australia to receive coordinate geometry training in one of two ways: a conventional way and a modified way. After the training, the students were asked to solve a coordinate geometry problem. The time to complete the problem was recorded, but five students in the “conventional” group did not complete the solution in the five minute allotted time.

Variables:

- Time = time (in seconds) student worked on problem
- Treatment = Modified or Conventional
- Censor = 1 if the individual did not complete the problem in 5 minutes, 0 if they did

Issue with Usual Analysis

Compare times to complete problems between two groups.

Method: Inference for $\mu_1 - \mu_2$: requires calculation of sample means and sample standard deviations for both groups.

Cannot calculate sample mean and standard deviation for Conventional group since five students in this group did not complete the puzzle.

Data Description and Variables

Food safety experts studied different ways to eradicate E coli from hamburger. In a study, six samples were exposed to a Control Treatment and 6 samples were exposed to an Active Treatment. After the treatment, the number of E coli in the sample both a Control and Active Treatment, the number of E coli in the samples was determined.

Variables:

- Number of E coli (in colony-forming units per gram)
- Treatment = Control or Active

Issue with Usual Analysis

Compare times to complete problems between two groups.

Method: Inference for $\mu_1 - \mu_2$: based on sample means and sample standard deviations for both groups.

Observations:

- Control: 0.17, 0.35, 0.22, 2.5, 0.53, 0.57
- Active: 0.11, 0.19, 0.03, 0.18, 0.02, 0.11

Very large observation in Control group. Will greatly affect sample mean and standard deviation for Control group.

Non-Parametric Methods

Two Samples:

- Wilcoxon Rank Sum Test

Paired Data:

- Sign Test
- Wilcoxon Signed Rank Test